

# Rank-Based Action Parameterisation for Multi-Objective Joint Day-Ahead and Intraday Battery Storage Dispatch via Deep Reinforcement Learning

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## ABSTRACT

Joint Day-Ahead (DA) and real-time Intraday (ID) dispatch for grid-scale battery energy storage systems (BESS) requires irrevocable forward commitments and real-time power rebalancing under price uncertainty, degradation, and carbon externalities. This is a high-dimensional stochastic control problem where standard approaches—rolling-horizon linear programming (LP) and reinforcement learning (RL)—are inadequate. This work introduces a three-dimensional action parameterisation that reduces joint DA–ID scheduling to dispatch power, commitment aggressiveness, and planning power, with a rank-based expansion algorithm mapping these to a state-of-charge-feasible 48-slot schedule. Applied with Soft Actor-Critic (SAC) and benchmarked against Dueling Double DQN (D3QN), Double DQN (DDQN), and DQN baselines plus an LP oracle over a 219-day GB test (five seeds), SAC achieves 42.2% LP-oracle efficiency (vs 31.6% D3QN). A Pareto sweep reveals a non-monotonic optimum at a carbon penalty weight of 0.1 where profit and carbon displacement are jointly maximised. An evaluation-time action-discretisation ablation confirms that the £84k SAC–D3QN profit advantage is entirely attributable to the rank-based parameterisation and the policy it enabled: snapping SAC’s continuous actions to the same 11-level discrete granularity as D3QN leaves performance unchanged, ruling out continuous precision as a confound.

**Keywords:** battery energy storage, deep reinforcement learning, action space design, UK electricity market, day-ahead scheduling, soft actor-critic

## NOMENCLATURE

|                        |                                                                               |                    |                                                           |
|------------------------|-------------------------------------------------------------------------------|--------------------|-----------------------------------------------------------|
| $\alpha$               | DA commitment aggressiveness (fraction, $\in [0, 1]$ )                        | $N_s$              | Number of cells in series per cabinet                     |
| CI                     | Grid average emission factor / carbon intensity ( $\text{gCO}_2/\text{kWh}$ ) | $P_t$              | Applied power (MW); $P > 0$ = discharge, $P < 0$ = charge |
| $\Delta t$             | Settlement period duration (0.5 h)                                            | $P_{\max}$         | Maximum power rating (MW)                                 |
| $E_{\max}$             | Usable energy capacity (MWh)                                                  | $p_t^{\text{DA}}$  | Day-Ahead price (£/MWh)                                   |
| $\eta_{\text{ch/dis}}$ | Coulombic charging/discharging efficiency (fraction)                          | $p_t^{\text{ID}}$  | Intraday price (£/MWh)                                    |
| $\kappa$               | Marginal degradation cost (£/MWh)                                             | $p_t^{\text{UKA}}$ | UK Allowance carbon price (£/tCO <sub>2</sub> )           |
|                        |                                                                               | $Q_{\text{rated}}$ | Rated charge capacity (Ah)                                |
|                        |                                                                               | $R_{\text{sys}}$   | System internal resistance ( $\Omega$ )                   |
|                        |                                                                               | $\zeta_t$          | State of charge (SoC) at time $t$ (fraction)              |
|                        |                                                                               | $V_{\text{OCV}}$   | Open-circuit voltage (V)                                  |

## 1 Introduction

Battery Energy Storage Systems (BESS) are being deployed at scale across Great Britain (GB) to integrate variable renewable generation under the UK’s net-zero commitment [1], with approximately 7.2 GW (11.8 GWh) operational as of 2026 [2]. The primary revenue stream is wholesale energy arbitrage across the Day-Ahead (DA) and Intraday (ID) markets ( $\sim 50\%$  of

the 2024 GB BESS revenue stack [3]), where operators commit forward positions and rebalance in real time under price uncertainty [4, 5]. Figure 1 illustrates the environment modelled in this paper.

Profit-maximising dispatch alone is insufficient. Aggressive cycling accelerates capacity fade and erodes long-term asset value [6]. Arbitrage can also increase system emissions: BESS may charge during carbon-

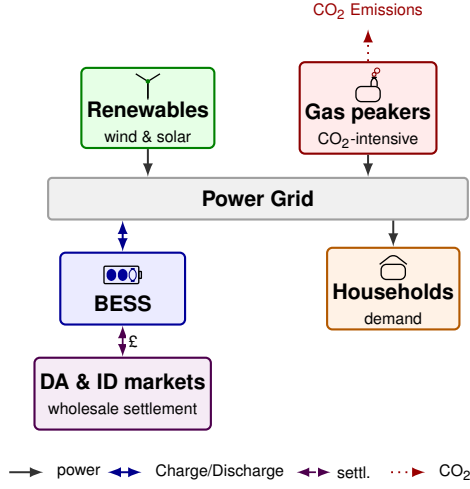


Figure 1: Environment model. BESS settles through DA and ID markets and displaces gas-peaker emissions at peak demand.

intensive periods and discharge when marginal generation is cleaner. Mixed-integer studies confirm that profit-optimal arbitrage can significantly raise system CO<sub>2</sub>, while modest multi-objective adjustments largely eliminate this effect [7]. The UK ETS, at recent carbon prices of approximately £40–50/tCO<sub>2</sub> [8], penalises carbon-intensive dispatch. BESS operation is therefore a multi-objective control problem requiring simultaneous balancing of profit, degradation, and carbon impact.

Optimisation-based approaches—rolling-horizon LP and model predictive control (MPC)—are the established tools for BESS scheduling in wholesale markets [4], but they require accurate price forecasts and struggle with non-convexities from SoC-dependent efficiency, throughput degradation, and irrevocable DA commitments. Reinforcement learning (RL) offers a model-free alternative: an agent learns a dispatch policy through direct interaction with a simulated market environment, eliminating the forecast dependency entirely [9]. Deep RL has demonstrated strong results in single-market BESS arbitrage [10, 11] and PV-BESS self-scheduling [12], but joint DA-ID dispatch under realistic battery physics and explicit multi-objective trade-offs remains underexplored.

Most RL-BESS studies use single-market discrete actions with simplified battery models [10, 11]; the few continuous-action approaches [11, 12] do not address DA commitment scheduling. Formulating joint DA-ID control with one output per half-hour slot plus real-time dispatch yields a 49-dimensional problem. Preliminary experiments applying Soft Actor-Critic (SAC) [13] directly to this formulation suffered entropy collapse and

converged to an idle policy ( $\alpha_{\text{ent}} \approx 0.006$ , idle fraction >91% within 90 episodes; Section 3.3.2), motivating a structured parameterisation that exploits DA price ranks. Three value-based baselines from the Deep Q-Network (DQN) family—DQN, Double DQN (DDQN), and Dueling Double DQN (D3QN)—serve as comparators (Section 4.1).

This work makes three contributions:

- (i) A 3D continuous action-space parameterisation for SAC that enables joint DA-ID dispatch without slot-wise control, using price-percentile ranking and forward SoC feasibility checking (Section 3.3.2).
- (ii) A BESS simulation environment integrating R-int ECM physics, throughput degradation, and dual-market settlement with UKA carbon pricing (Section 3.1).
- (iii) A comparative evaluation—including an evaluation-time action-discretisation ablation that attributes  $\approx 100\%$  of the SAC-D3QN gap to the rank-based parameterisation rather than continuous precision—and a Pareto sweep showing that a moderate carbon penalty simultaneously improves profit and carbon displacement (Sections 5).

## 2 Literature Review

This review covers three strands of prior work relevant to the problem formulation of this report: deep reinforcement learning for BESS dispatch, structured action spaces for high-dimensional scheduling, and multi-objective reward design with carbon internalisation.

Cao et al. [10] demonstrated profitable GB wholesale arbitrage using a noisy-network DQN with CNN-LSTM price forecasts and cycle-based degradation; however, the forecasting module introduces a fragile pipeline that must be retrained across market regimes, whereas the RL policy itself could in principle adapt through continued interaction. Karimi-Madahi et al. [11] apply distributional SAC and DQN to risk-sensitive arbitrage in imbalance settlement with explicit cycle constraints, but restrict the action space to single-market dispatch levels and so sidestep the combinatorial challenge of joint DA-ID scheduling. Perez et al. [12] integrate Seq2Seq forecasting with PPO and SAC for price-based self-scheduling of utility-scale PV-BESS, but similarly do not model DA commitment structure.

Preference-based multi-objective RL has been explored in microgrid settings [14], though at a scale that does not transfer directly to wholesale BESS. Across these four studies the shared gap is a faithful DA–ID formulation, which requires 48 half-hour commitments plus a real-time dispatch decision at each step — a 49-dimensional action space that no prior RL-BESS study reviewed here has attempted to solve end-to-end.

The dimensionality challenge of joint DA–ID formulation motivates structured continuous action spaces. SAC [13] provides a foundation for continuous control, but this 49-dimensional DA–ID formulation produces sparse delayed rewards and empirical entropy collapse [15]. Parameterised action spaces [16] address this by mapping low-dimensional continuous parameters into structured high-dimensional sequences. In related domains, two-stage frameworks separate day-ahead scheduling from real-time redispatch in networked microgrids [17], and deep RL has been applied to real-time microgrid energy management with continuous actions under uncertainty [18]. However, these approaches either use model-based optimisation for the day-ahead layer or focus purely on real-time dispatch; none provide an end-to-end DRL formulation jointly learning DA commitments and ID redispatch under realistic BESS physics.

A further design requirement for RL-based BESS control is Markov-compatible degradation accounting. In RL, the agent learns from individual transitions—each recording the current state, action taken, and reward received—so every reward signal must be computable from the current step alone, without access to prior history. Standard Rainflow cycle-counting violates this by accumulating damage over the full charge–discharge history. Cortés-Arcos et al. [6] resolve this with a throughput-based model in which degradation cost is proportional to per-step energy throughput, making it fully compatible with RL training. Keske et al. [7] further show via MILP that profit-optimal arbitrage raises system emissions, though their approach requires price foresight and cannot be deployed reactively. Weighted linear scalarisation [19] provides a tractable multi-objective framework, though it recovers only the convex hull of the Pareto front—a limitation that becomes relevant when the trade-off surface has non-convex structure, as observed in this work.

Three insights emerge from reviewing these strands. First, the persistent reliance on external price forecast-

ing pipelines [10, 12] reveals that no existing RL-BESS policy has been evaluated in a fully model-free end-to-end setting for joint DA–ID dispatch; the forecasting module is both a practical fragility and a scientific confound. Second, the action-space complexity of a 48-slot DA schedule is a structural barrier that has not been resolved by parameterised-action or two-stage approaches [16, 17], because those frameworks either use model-based DA layers or restrict dispatch to a single market. Third, the carbon-arbitrage tension identified by Keske et al. [7] has only been studied under perfect foresight via MILP; whether a reactive RL policy can achieve favourable profit–carbon trade-offs without foresight remains an open question. These three insights motivate contributions (i)–(iii) outlined in Section 1.

### 3 Problem Formulation

The problem is to learn a stochastic policy  $\pi_\theta(a_t | s_t)$  that jointly schedules Day-Ahead commitments and real-time Intraday dispatch for a grid-scale BESS, maximising the expected discounted return subject to state-of-charge and power constraints:

$$\begin{aligned} \pi^* \in \arg \max_{\pi} \mathbb{E}_{\pi} \left[ \sum_{t=0}^{T-1} \gamma^t r_t \right] \\ \text{s.t. } \zeta_t \in [\zeta_{\min}, \zeta_{\max}], |P_t| \leq P_{\max} \end{aligned} \quad (1)$$

where  $r_t$  combines revenue, degradation, and carbon terms (Section 3.3.4). Figure 2 illustrates how the SAC agent observes market state, selects a 3D action, and receives reward  $r_t$  from the BESS environment.

#### 3.1 System Specification & Battery Physics

The system is modelled as a deterministic R-int equivalent circuit of the CATL EnerOne LFP cabinet (1P416S topology, specifications in Table 1) [20], scaled to 268 parallel cabinets matching the Minety Project (99.9 MW/99.8 MWh) [21, 22].

The R-int ECM dynamically computes  $I^2R$  ohmic losses; accordingly, the charging efficiency  $\eta_{\text{ch}} = 0.995$  and discharging efficiency  $\eta_{\text{dis}} = 1.0$  represent purely Coulombic efficiency, consistent with ECM-based optimisation practice [23, 24]. The pack OCV is linearly interpolated from the manufacturer’s cell table [20] (reproduced in Appendix A) and scaled by series topology:

$$V_{\text{OCV,pack}}(\zeta_t) = N_s \times V_{\text{cell}}(\zeta_t) \quad (2)$$

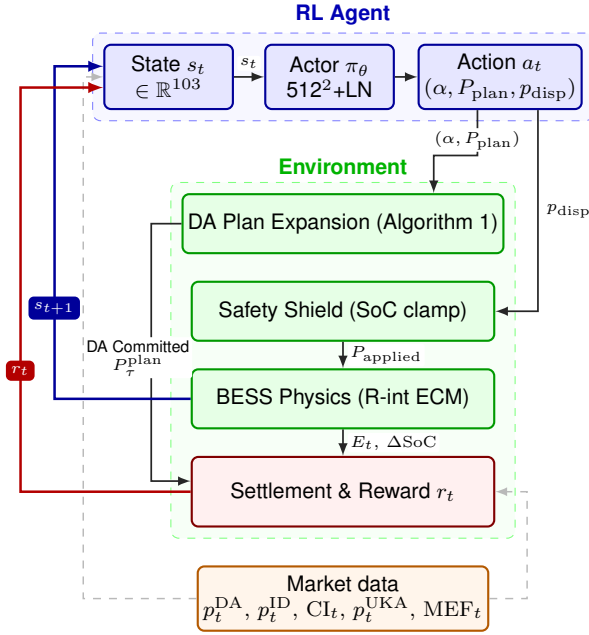


Figure 2: System architecture. The actor’s action  $(\alpha, P_{\text{plan}}, p_{\text{disp}})$  feeds DA Plan Expansion (Algorithm 1) and the Safety Shield; Financial Settlement returns the normalised reward  $r_t$ .

Table 1: BESS hardware specification (CATL EnerOne LFP [lithium iron phosphate], 268 cabinets).

| Parameter                                 | Value                                         |
|-------------------------------------------|-----------------------------------------------|
| Cell chemistry [20]                       | LFP (CATL 280 Ah)                             |
| Topology per cabinet [20]                 | 1P416S                                        |
| No. cabinets (Derived)                    | 268 (parallel)                                |
| Nominal energy per cabinet [20]           | 372.7 kWh / cabinet                           |
| System energy $E_{\text{max}}$ (Derived)  | 99.9 MWh                                      |
| Power rating $P_{\text{max}}$ (0.5C [20]) | 49.95 MW                                      |
| SoC limits [20]                           | [0.10, 0.90]                                  |
| Charge efficiency [23]                    | $\eta_{\text{ch}} = 0.995$                    |
| Discharge efficiency [23]                 | $\eta_{\text{dis}} = 1.00$                    |
| System resistance (Derived [20])          | $R_{\text{sys}} = 6.20 \times 10^{-4} \Omega$ |
| Degradation cost [6]                      | $\kappa = 10.0 \text{ £/MWh}$                 |

The cyclic throughput degradation cost follows Cortés-Arcos et al. [6]:  $\kappa = C_{\text{bat}}/Q_{\text{lifetime}} \approx \text{£}10/\text{MWh}$ , using a conservative UK/Europe pack cost of  $\text{£}95\text{--}103/\text{kWh}$ , derived from the China volume-weighted price of  $\text{£}84/\text{kWh}$  and a 44–56% European premium ( $\text{USD}/\text{GBP} \approx 1.27$ ), all per the BNEF 2025 Battery Price Survey [25]; 6,000-cycle lifetime per [20]; full derivation in Appendix A.

## 3.2 Market Structure & Settlement

The BESS operates in the UK wholesale electricity market ( $P_t > 0$  discharge,  $P_t < 0$  charge;  $P_{\text{max}} \approx 49.95\text{MW}$  at 0.5C [20]), with revenue from DA forward settlement, ID rebalancing at the Market Index

Data (MID) price, and carbon-aware dispatch penalised at UK Allowance (UKA) prices. DA commitments are irrevocable; deviations settle via ID only. The safety shield (Section 3.3.3) prevents SoC violations, so SSP/SBP imbalance penalties are not modelled. The operator is a price-taker with no market impact.

The environment operates at one-step resolution of one Settlement Period (SP,  $\Delta t = 0.5\text{h}$ ), with delivery index  $\tau_t \in \{1, \dots, 48\}$  indexing the current half-hour slot. The next day’s DA prices  $\mathbf{p}^{\text{DA}}$  and carbon intensity forecasts  $\mathbf{CI}^{\text{DA}}$  are revealed only when trade time  $\bar{\tau} \geq T_{\text{publish}}$  (12:00 GMT publication time, on trade day  $D-1$ ), signalled by binary flag  $f_{\text{DA},t}$ .

## 3.3 Markov Decision Process Formulation

The optimisation in Eq. (1) is instantiated as a discounted Markov Decision Process (MDP)  $(\mathcal{S}, \mathcal{A}, \mathcal{T}, \mathcal{R}, \gamma)$  with episode horizon  $T = 48 \times \text{episode\_days}$  half-hour steps, where  $\gamma \in (0, 1]$  is the discount factor; operationally, this corresponds to maximising net arbitrage profit while internalising degradation and carbon costs and managing SoC inter-temporally to preserve future trading optionality.

### 3.3.1 State Space

At each timestep  $t$ , the agent observes seven scalar signals and two 48-dimensional look-ahead vectors: SoC  $\zeta_t$ ; realised intraday price  $p_t^{\text{ID}}$ ; realised grid carbon intensity  $\text{CI}_t$ ; within-day settlement index  $\tau_t$ ; the previous applied power  $P_{t-1}$ ; the DA publication flag  $f_{\text{DA},t}$ ; the planned power commitment  $P_t^{\text{plan}}$ ; the forecast DA price curve  $\mathbf{p}_{48}^{\text{DA}}$  for the next delivery day; and the corresponding forecast CI curve  $\mathbf{CI}_{48}^{\text{DA}}$ . These assemble into the 103-dimensional state vector:

$$s_t = \left( \underbrace{\zeta_t}_{\text{SoC}}, \underbrace{p_t^{\text{ID}}}_{\text{price}}, \underbrace{\text{CI}_t}_{\text{carbon intensity}}, \underbrace{\tau_t}_{\text{time}}, \underbrace{P_{t-1}}_{\text{prev}}, \right. \\ \left. \underbrace{f_{\text{DA}}}_{\text{flag}}, \underbrace{P_t^{\text{plan}}}_{\text{plan}}, \underbrace{\mathbf{p}_{48}^{\text{DA}}}_{\text{DA curve}}, \underbrace{\mathbf{CI}_{48}^{\text{DA}}}_{\text{CI curve}} \right) \quad (3)$$

summarised in Table 2.

The DA curve and CI forecast are zero-masked when  $f_{\text{DA},t} = 0$ , preventing information leakage.  $S_{\text{price}}$  and  $S_{\text{CI}}$  are 95th-percentile values on the training split, following percentile-based normalisation [26] to avoid the signal compression that max-based scaling suffers under extreme outliers.

Table 2: State vector  $s_t \in \mathbb{R}^{103}$  observed at each half-hour step.

| Feature                                 | Normalisation                 | Dim        |
|-----------------------------------------|-------------------------------|------------|
| SoC $\zeta_t$                           | Raw, clipped to $[0.1, 0.9]$  | 1          |
| ID price $p_t^{\text{ID}}$              | $p/S_{\text{price}}$          | 1          |
| Carbon intensity $\text{CI}_t$          | $\text{CI}/S_{\text{CI}}$     | 1          |
| Settlement period $\tau_t$              | $\tau/48$                     | 1          |
| Prev. power $P_{t-1}$                   | $P/P_{\text{max}}$            | 1          |
| DA flag $f_{\text{DA},t}$               | $\{0, 1\}$                    | 1          |
| Planned power $P_t^{\text{plan}}$       | $P/P_{\text{max}}$            | 1          |
| DA price curve $\mathbf{p}^{\text{DA}}$ | $\mathbf{p}/S_{\text{price}}$ | 48         |
| CI forecast $\mathbf{CI}^{\text{DA}}$   | $\mathbf{CI}/S_{\text{CI}}$   | 48         |
| <b>Total</b>                            |                               | <b>103</b> |

### 3.3.2 Action Space

Two action space formulations are implemented: a discrete space for value-based agents and a continuous parameterised space for policy-gradient agents.

**Discrete Formulation.** The discrete action at each step is:

$$a_t = \begin{pmatrix} a_t^{\text{dispatch}} \\ a_t^{\text{plan}} \\ k_t \end{pmatrix} \quad (4)$$

where:

1.  $a_t^{\text{dispatch}} \in \mathcal{P}$ : real-time dispatch power, from 11 levels between  $-P_{\text{max}}$  and  $+P_{\text{max}}$  (0 for idle).
2.  $a_t^{\text{plan}} \in \mathcal{P}$ : power volume to commit to the next day’s DA schedule.
3.  $k_t \in \{0, \dots, 47\}$ : half-hour slot of the next day’s schedule to update with  $a_t^{\text{plan}}$ .

Planning actions apply only when  $f_{\text{DA},t} = 1$  and the next day’s delivery day exists; from DA price publication at 12:00, the agent has 24 steps to write one DA slot per step.

**Total discrete action space:**  $|\mathcal{A}| = 11 \times 11 \times 48 = 5,808$  actions. At this scale, two learning pathologies are well documented in the DRL literature [15, 27]: (i) *credit-assignment sparsity*, because only one of 5,808 actions is sampled per step, making reward signals diffuse across the Q-table; and (ii) *Q-value overestimation* [28], which is amplified in large discrete spaces because the max operator in the Bellman target selects biased estimates from a larger candidate set. Both motivate the DDQN and D3QN choices in Section 4.1, and Section 5.1 provides empirical confirmation: the DQN

$\rightarrow \text{DDQN} \rightarrow \text{D3QN}$  hierarchy recovers the uplifts expected if each pathology binds.

**Continuous Parameterised Formulation.** A DA schedule in an arbitrage context is structurally determined by *which* half-hours to activate and *how much* power to commit to them—two latent parameters that replace the 48 coupled slot-wise outputs of the direct formulation. Adding real-time dispatch intensity as a third dimension yields a 3D action space in which  $\alpha$  controls slot activation,  $P_{\text{plan}}$  sets the committed power level, and  $a_t^{\text{dispatch}}$  handles intraday rebalancing—a domain-specific reduction, not a general dimensionality claim.

The SAC actor outputs three continuous variables:

$$a_t = \begin{pmatrix} a_t^{\text{dispatch}} \\ a_t^{\text{pwr}} \\ a_t^{\text{agg}} \end{pmatrix} \in [-1, 1]^3 \quad (5)$$

mapped to physical quantities via:

$$P_{\text{dispatch}} = a_t^{\text{dispatch}} \cdot P_{\text{max}} \quad (6)$$

$$P_{\text{plan}} = \frac{a_t^{\text{pwr}} + 1}{2} \cdot P_{\text{max}} \quad (7)$$

$$\alpha = \frac{a_t^{\text{agg}} + 1}{2} \in [0, 1] \quad (8)$$

The aggressiveness  $\alpha$  and planning power  $P_{\text{plan}}$  are expanded into a 48-slot DA schedule by Algorithm 1. The algorithm ranks the 48 DA prices by percentile, then performs a single chronological pass that discharges at rank  $> 1 - \alpha/2$  and charges at rank  $< \alpha/2$ , rejecting slots whose forward-simulated SoC would leave the allowed band. The forward  $\delta$  uses ideal Coulomb counting as an optimistic feasibility bound; the realised transition (Eq. 9) applies  $\eta_{\text{ch}}$  in real-time settlement. The resulting schedule is SoC-feasible by construction, bypassing the need for the actor to jointly learn feasibility and profitability across 48 coupled outputs.

To reflect DA contract irrevocability, a **plan latching** mechanism commits the DA schedule on the first step where  $f_{\text{DA},t} = 1$  (12:00 GMT trade day  $D-1$ ), and holds until day rollover ( $\tau = 48 \rightarrow 1$ ); this prevents within-day switching between aggressiveness levels. Discrete agents construct their plan slot-by-slot during the post-publication window under the same constraint. Figure 3 shows a real SAC step-by-step use of Algorithm 1.

**Algorithm 1** Rank-Based DA Plan Expansion

**Require:** Next day's DA price curve  $\mathbf{p}^{\text{DA}} \in \mathbb{R}^{48}$ ; aggressiveness  $\alpha \in [0, 1]$ ; planning power  $P_{\text{plan}} \in [0, P_{\text{max}}]$ ; current  $\zeta$

**Ensure:** SoC-feasible 48-slot schedule  $\mathbf{P}^{\text{DA}} \in \mathbb{R}^{48}$

```

1: Compute percentile ranks  $\rho_s \in [0, 1]$  for each slot  $s$ 
2:  $\zeta_{\text{sim}} \leftarrow \zeta$   $\triangleright$  Forward simulation
3: for  $s = 0, 1, \dots, 47$  do  $\triangleright$  Chronological pass
4:   if  $\rho_s > 1 - \alpha/2$  then  $\triangleright$  Expensive slot:
     discharge
5:      $\delta \leftarrow P_{\text{plan}} \cdot \Delta t / E_{\text{max}}$ 
6:     if  $\zeta_{\text{sim}} - \delta \geq \zeta_{\text{min}}$  then
7:        $P_s^{\text{DA}} \leftarrow +P_{\text{plan}}$ ;  $\zeta_{\text{sim}} \leftarrow \zeta_{\text{sim}} - \delta$ 
8:     end if
9:   else if  $\rho_s < \alpha/2$  then  $\triangleright$  Cheap slot: charge
10:     $\delta \leftarrow P_{\text{plan}} \cdot \Delta t / E_{\text{max}}$ 
11:    if  $\zeta_{\text{sim}} + \delta \leq \zeta_{\text{max}}$  then
12:       $P_s^{\text{DA}} \leftarrow -P_{\text{plan}}$ ;  $\zeta_{\text{sim}} \leftarrow \zeta_{\text{sim}} + \delta$ 
13:    end if
14:   else
15:      $P_s^{\text{DA}} \leftarrow 0$   $\triangleright$  Inactive slot
16:   end if
17: end for
18: return  $\mathbf{P}^{\text{DA}}$ 

```

**Properties and limitations.** The algorithm is *price-rank invariant*: activation depends on percentile ranks, making it robust to uniform price-level shifts. Two constraints limit expressivity—symmetric activation and uniform  $P_{\text{plan}}$  across active slots—chosen to prevent entropy collapse (Section 1); relaxing them is left for future work (Section 5.5).

### 3.3.3 State Transition

The R-int circuit equation  $P_t = I_t(V_{\text{OCV}}(\zeta_t) - I_t R_{\text{sys}})$  is solved for the physically valid root; a safety shield then clamps  $I_t$  so that SoC stays within  $[\zeta_{\text{min}}, \zeta_{\text{max}}]$  without imbalance settlement penalties. The sign convention is  $I_t > 0$  for discharge and  $I_t < 0$  for charge, consistent with  $P_t > 0$  denoting power export to the grid. The SoC advances by Coulomb counting with asymmetric Coulombic efficiency (self-discharge  $< 0.5\% \text{ yr}^{-1}$ , negligible at 30-min resolution [20]):

$$\zeta_{t+1} = \begin{cases} \zeta_t - \eta_{\text{ch}} \cdot \frac{I_t \cdot \Delta t}{Q_{\text{rated}}} & P_t < 0 \\ \zeta_t - \frac{1}{\eta_{\text{dis}}} \cdot \frac{I_t \cdot \Delta t}{Q_{\text{rated}}} & P_t > 0 \\ \zeta_t & P_t = 0 \end{cases} \quad (9)$$

followed by hard clamping to  $[\zeta_{\text{min}}, \zeta_{\text{max}}]$  per [20]. DA plans promote to the active schedule at day rollover ( $\tau =$

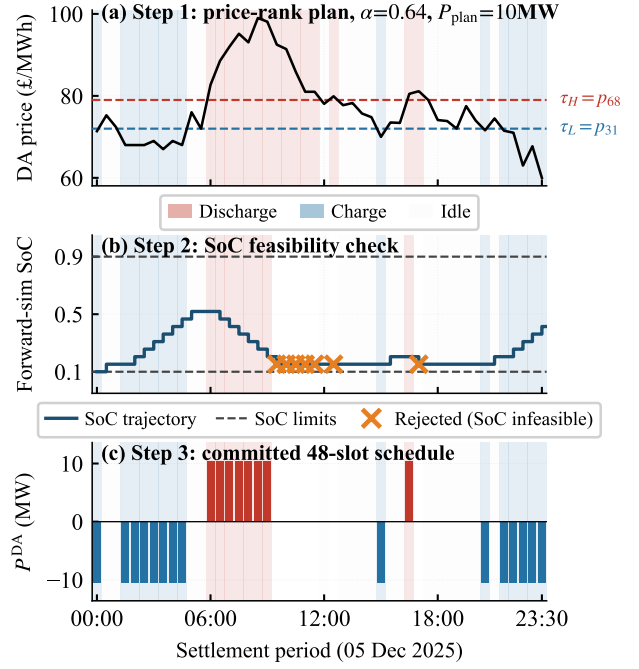


Figure 3: Algorithm 1 on a representative SAC rollout: (a) price-rank classification, (b) forward SoC simulation, (c) committed 48-slot schedule.

48  $\rightarrow$  1): discrete agents write one slot per step during the 24-step post-publication window; continuous agents write the full plan atomically via Algorithm 1. Per-step financial settlement then computes:

$$R_{\text{DA}} = P_t^{\text{plan}} \cdot \Delta t \cdot p_t^{\text{DA}} \quad (10)$$

$$R_{\text{ID}} = (P_t^{\text{act}} - P_t^{\text{plan}}) \cdot \Delta t \cdot p_t^{\text{ID}} \quad (11)$$

$$C_{\text{deg}} = \kappa \cdot |P_t^{\text{act}}| \cdot \Delta t \quad (12)$$

$$C_{\text{carbon}} = p_t^{\text{UKA}} \cdot \frac{E_{\text{imp}} \cdot \text{CI}_t - E_{\text{exp}} \cdot \text{MEF}_t}{10^6} \quad (13)$$

where  $E_{\text{imp/exp}} = \max(\mp P_t^{\text{act}} \cdot \Delta t, 0) \times 10^3 \text{ kWh}$ ; imported energy is costed at the average emission factor (AEF)  $\text{CI}_t$ , while exported energy uses the marginal emission factor  $\text{MEF}_t$ , since discharging displaces the marginal plant rather than the average grid mix [29]. Market prices advance from the historical dataset to complete the transition.

### 3.3.4 Reward Function

The net cashflow per step is:

$$R_t^{\text{net}} = R_{\text{DA}} + R_{\text{ID}} - C_{\text{deg}} - \lambda_{\text{CI}} \cdot C_{\text{carbon}} \quad (14)$$

The scalar training reward applies linear normalisation:

$$r_t = \frac{R_t^{\text{net}} / E_{\text{max}}}{S_{\text{universal}}} \quad (15)$$

where  $S_{\text{universal}} = S_{\text{profit}}/E_{\text{max}}$  and  $S_{\text{profit}}$  is the 95th percentile of per-step profit magnitude from the training split.

Early experiments with  $\tanh(R/S)$  suppressed gradients for  $|R/S| > 1$ , rendering degradation and carbon signals effectively invisible. Linear normalisation maps typical step rewards to  $[-0.2, +0.2]$  while preserving relative magnitudes, so cost signals propagate to the policy.

## 4 Experimental Set-up

### 4.1 Discrete Value-Based Agents

Three value-based agents operate on the flat 5,808-action discrete space: DQN (baseline), DDQN (overestimation fix), and D3QN (best discrete agent, primary comparator to SAC). All share the same target equation structure and training infrastructure; architectural differences are described below.

**DDQN.** Standard DQN suffers from systematic Q-value overestimation because the same network both selects and evaluates the greedy action in the Bellman target. DDQN [28] corrects this by decoupling action selection (online network  $Q_{\text{eval}}$ ) from action evaluation (target network  $Q_{\text{target}}$ ):

$$y_t = r_t + \gamma Q_{\text{target}}\left(s_{t+1}, \arg \max_{a'} Q_{\text{eval}}(s_{t+1}, a')\right) \cdot (1 - \mathbf{1}[\text{done}]) \quad (16)$$

The shared Q-network maps the 103-dimensional observation through two hidden layers (256 units, ReLU) to Q-values for all 5,808 actions. Huber loss is minimised with gradient norms clipped at 10.0;  $Q_{\text{target}}$  is hard-synchronised every 15,000 steps.

**D3QN (Dueling DDQN).** D3QN [27] replaces the single output head with two parallel streams: a state-value stream  $V(s)$ , estimating the value of being in state  $s$  regardless of action, and an advantage stream  $A(s, a)$ , capturing the relative benefit of each action. These combine to factorise the Q-value:

$$Q(s, a) = V(s) + \left( A(s, a) - \frac{1}{|\mathcal{A}|} \sum_{a'} A(s, a') \right) \quad (17)$$

This decomposition is critical at 5,808-action scale: when market spreads are compressed,  $V(s)$  propagates a “low-value state” signal to all 5,808 actions in a single Bellman update, solving the credit-assignment sparsity problem that degrades DDQN in large discrete spaces. All three discrete agents were independently tuned via Bayesian hyperparameter optimisation (HPO) using Optuna TPE [30] over 30 trials each; the resulting configurations are given in the supplementary appendix (Tables C1–C3).

### 4.2 Soft Actor-Critic Agent

Soft Actor-Critic (SAC) [13] operates on the 3-dimensional continuous action space defined in Section 3.3.2, maximising the entropy-regularised objective:

$$\pi^* = \arg \max_{\pi} \mathbb{E}_{\pi} \left[ \sum_{t=0}^{T-1} \gamma^t (r_t + \alpha_{\text{ent}} \mathcal{H}(\pi(\cdot | s_t))) \right] \quad (18)$$

where  $\alpha_{\text{ent}}$  is the automatically tuned entropy temperature [13] and  $\mathcal{H}$  is the policy entropy.

Three implementation-specific choices distinguish this SAC from the reference algorithm. First, Layer Normalisation [31] is applied after each hidden layer in both the actor ( $103 \rightarrow 512 \rightarrow 512 \rightarrow (\mu, \log \sigma)$ ) and twin critics ( $106 \rightarrow 512 \rightarrow 512 \rightarrow 1$ ) to stabilise gradients across the heterogeneous 103-dimensional state (prices in £/MWh, SoC in  $[0, 1]$ , carbon intensity in gCO<sub>2</sub>/kWh). Second, the critic Bellman target incorporates a reward scale  $\omega = 7$ :

$$y_t = r_t \cdot \omega + (1 - \mathbf{1}[\text{done}]) \cdot \gamma \left( \min_{i=1,2} Q_{\theta'_i}(s_{t+1}, \tilde{a}_{t+1}) - \alpha_{\text{ent}} \log \pi(\tilde{a}_{t+1} | s_{t+1}) \right) \quad (19)$$

found via Optuna HPO to compensate for the small per-step reward magnitudes produced by the linear normalisation in Section 3.3.4. Third, a 5,000-step uniform warmup populates the replay buffer before learning begins, preventing early critic updates from overfitting to a narrow initial state distribution. Full architectural details and hyperparameters are given in Appendix B.

### 4.3 Training Protocol

Market and environmental data cover 2022-01-01 to 2026-01-01 at 30-minute resolution. Day-Ahead prices



and ICE UK Allowance futures (UKAFMC1) are sourced from LSEG Refinitiv Eikon; Intraday MID prices from Elexon BMRS (N2EX and APX); national carbon intensity from the NESO API; and the marginal emission factor is computed from Elexon FUELHH generation data using a ramp-based heuristic over CCGT, coal, and oil plant.

Training uses a three-phase chronological split (Table 3) to avoid temporal information leakage;  $k$ -fold cross-validation would introduce look-ahead bias in financial time-series RL [32]. The training window deliberately spans the 2022 post-invasion crisis to late 2024 to expose the agent to extreme conditions, yielding 48,912 timestep observations ( $1,019 \text{ days} \times 48 \text{ SPs}$ ). Reward-normalisation constants are computed on data from 2023 onwards, excluding the 2022 crisis outliers ( $> \text{£}4,000/\text{MWh}$ ) which would otherwise compress post-crisis rewards and suppress gradients.

Table 3: Chronological data split (70/15/15).

| Split | Date Range              | Days  | Purpose         |
|-------|-------------------------|-------|-----------------|
| Train | 2022-01-01 – 2024-10-18 | ~1019 | Policy learning |
| Val   | 2024-10-19 – 2025-05-25 | ~218  | HPO + selection |
| Test  | 2025-05-26 – 2026-01-01 | ~219  | Final eval.     |

Each agent is trained for 1,500 episodes on the training split. One episode spans 7 delivery days with randomised start dates and initial SoC  $\in [0.3, 0.7]$ ; episodes terminate on day completion and bootstrap targets for truncated episodes are preserved (not zeroed) to avoid Bellman underestimation at episode boundaries [9].

Every 50 episodes, a deterministic validation rollout is executed ( $\varepsilon = 0$  for D3QN; policy mean  $\tanh(\mu)$  for SAC); the best-scoring checkpoint is retained. No validation transitions enter the replay buffer. Early stopping triggers after 400 episodes without improvement (discrete agents: only after  $\varepsilon$ -decay completes). All results are reported as mean  $\pm$  standard deviation across five seeds  $\{0, 1, 2, 3, 4\}$ , with none withheld. Figure 4 shows SAC convergence across all five seeds.

#### 4.4 Baselines

Agent performance is benchmarked against four baselines:

1. **Idle** ( $P = 0$ ): Economic floor with zero revenue and zero degradation.
2. **Random (uniform from 5,808 actions)**: Expected

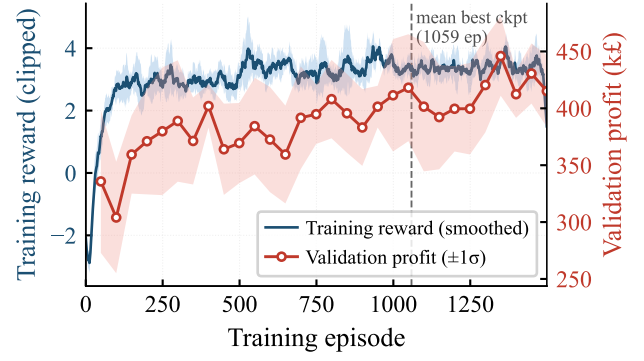


Figure 4: SAC training and validation performance across 5 seeds (mean  $\pm 1\sigma$ ). The dashed line indicates the mean optimal early-stopping checkpoint at 1,059 episodes.

strong negative profit due to degradation costs on random cycling.

3. **P20/P80 heuristic**: Charge below the 20th percentile of daily ID prices and discharge above the 80th percentile; idle otherwise. No DA commitments.
4. **LP Oracle (theoretical ceiling)**: Perfect-foresight linear programme maximising  $\sum_t (R_{\text{DA},t} + R_{\text{ID},t} - C_{\text{deg},t} - \lambda_{\text{CI}} \cdot C_{\text{carbon},t})$  subject to SoC dynamics, power limits, and SoC bounds, solved over the full 219-day test horizon using SciPy’s HiGHS solver.

#### 4.5 Evaluation Protocol

All agents are evaluated on the sequestered 219-day test split using a single continuous rollout ( $\zeta_0 = 0.5$ , no episodic resets) to prevent end-of-episode exploitation artefacts. The discrete agents use greedy selection ( $\varepsilon = 0$ ); SAC uses the deterministic policy mean  $\tanh(\mu)$  [13]. Step-level metrics are logged and key performance indicators (KPIs) computed post-hoc, reported as mean  $\pm$  standard deviation across the five seeds.

Five primary KPIs are reported: **Net profit** ( $R_{\text{DA}} + R_{\text{ID}} - C_{\text{deg}}$ , carbon term excluded); **EFC** ( $\sum_t |P_t| \Delta t / (2E_{\text{max}})$ , battery utilisation); **Rev/EFC** (profit per cycle, timing quality); **Win rate** (fraction of days with strictly positive profit); and **LP efficiency** (agent profit / oracle profit  $\times 100\%$ , scale-free comparator). Table 4 additionally decomposes DA/ID revenue share, degradation cost, idle fraction, DA plan utilisation, and carbon displacement to characterise *how* each agent achieves its result. To decompose the SAC–



D3QN profit gap into granularity and parameterisation contributions, a sixth configuration—SAC (disc-11)—evaluates the trained SAC policy with each continuous action dimension snapped to the nearest of 11 uniformly-spaced levels in  $[-1, 1]$  prior to the physical mapping (Eqs. 6–8); no retraining is performed.

## 5 Results

### 5.1 Aggregate Performance

The primary cross-formulation finding from Table 4 is that SAC (£335.0k  $\pm$  £45.8k, LP efficiency 42.2%) outperforms D3QN — the strongest discrete agent — by  $1.34\times$  (£250.9k  $\pm$  £24.6k, 31.6%). All RL agents decisively beat the P20/P80 heuristic (−£101k), confirming that static percentile rules cannot navigate the UK market. Table 4 also decomposes gross revenue by market. SAC derives 85.6% of gross revenue from DA (D3QN: 14.0%), with D3QN’s split inverted; DDQN’s DA commitments are net loss-making, with ID corrections rescuing profitability. DDQN’s negative DA share is the fingerprint of value overestimation propagating into poor DA commitments. Cross-seed dispersion on these shares is large ( $\sigma \approx 34$  pp for SAC/D3QN, 82 pp for DDQN); the figures characterise mean behaviour rather than every seed.

The DQN→DDQN→D3QN progression isolates algorithmic improvements within the discrete formulation. The SAC (disc-11) ablation further shows that  $\approx 100\%$  of the D3QN→SAC gap comes from the rank-based parameterisation, not continuous precision (full decomposition in Section 5.5).

Figure 5 reveals a pronounced “staircase” profit trajectory, with accumulation concentrated in Q4 2025, consistent with the seasonal demand uplift and gaspeaker entry (£60–100/MWh DA spikes) characteristic of autumn–winter transitions. All three RL agents spike contemporaneously at these high-spread events, confirming convergence to similar market-state representations despite architectural differences. SAC’s higher win rate (Table 4) is consistent with the rank-based parameterisation enabling more targeted activation of moderate-spread trading opportunities that the flat discrete formulation, with its slot-by-slot DA planning, fails to exploit; the disc-11 ablation confirms this is independent of action granularity.

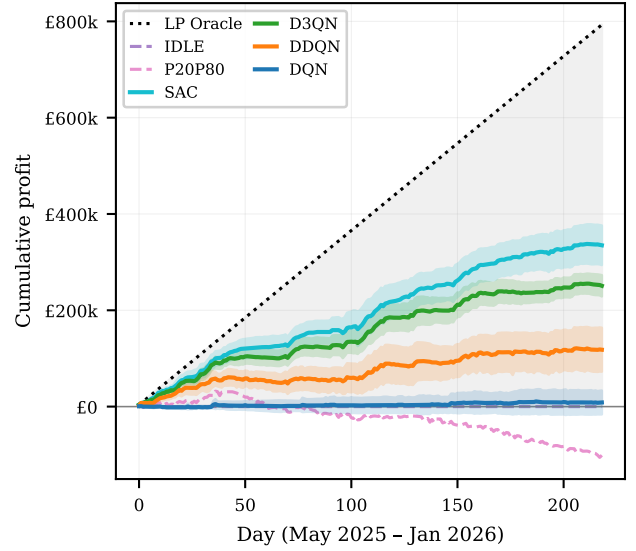


Figure 5: Cumulative net profit over the 219-day test period at  $\lambda_{CI} = 0.1$ . Shaded bands show  $\pm 1\sigma$  across five seeds; dotted line: LP oracle ceiling; dashed red line: P20/P80 heuristic.

### 5.2 Dispatch Behaviour

SAC utilises fewer DA slots than D3QN yet routes far more revenue through the forward market (Table 4), with mean absolute applied power 7.03 MW vs 4.30 MW respectively. This indicates a more selective, higher-intensity policy: the rank-based expansion concentrates dispatch into the highest-spread bands, avoiding marginal periods whose spread is insufficient to offset the £10/MWh degradation cost. D3QN commits more slots at lower intensity, pushing much of its revenue into ID corrections of over-planned DA positions.

Figure 6 illustrates the same pattern on 28 November 2025. Both agents exploit the overnight low-price window and a midday recharge, but SAC follows the price structure more precisely, restoring SoC earlier and enabling a cleaner second evening discharge. D3QN captures the same broad arbitrage structure but its over-planned DA positions require heavier ID correction, leaving less flexibility at the end of the day—a consequence of the flat discrete parameterisation rather than granularity, as confirmed by the disc-11 ablation (Section 5.5). The seven-day context is given in Appendix D; the heatmap in Figure 7 shows that SAC maintains concentrated early-morning and late-afternoon activation across the test set, whereas D3QN is more diffuse in time, consistent with its lower aggregate EFC and Rev/EFC.

Table 4: Test-set performance over 219 days (mean  $\pm$  std). SAC (disc-11) evaluates the trained continuous policy with actions snapped to 11 levels per dimension.

| Metric                                | SAC (3D)                           | SAC (disc-11)    | D3QN                | DDQN                               | DQN                              | P20/P80        | LP Oracle |
|---------------------------------------|------------------------------------|------------------|---------------------|------------------------------------|----------------------------------|----------------|-----------|
| <b>Block A: Financial Performance</b> |                                    |                  |                     |                                    |                                  |                |           |
| Profit (k£)                           | <b>335.0 <math>\pm</math> 45.8</b> | 335.5 $\pm$ 41.8 | 250.9 $\pm$ 24.6    | 118.1 $\pm$ 50.9                   | 8.6 $\pm$ 27.5                   | −101.1         | 794.1     |
| LP Efficiency (%)                     | <b>42.2 <math>\pm</math> 5.4</b>   | 42.3 $\pm$ 5.2   | 31.6 $\pm$ 2.9      | 14.9 $\pm$ 6.0                     | 1.1 $\pm$ 3.8                    | −12.7          | 100.0     |
| Win Rate (%)                          | <b>75.3 <math>\pm</math> 5.6</b>   | 75.3 $\pm$ 5.7   | 60.4 $\pm$ 2.5      | 52.1 $\pm$ 5.5                     | 49.5 $\pm$ 8.4                   | 48.9           | —         |
| EFC (cycles)                          | 180.3 $\pm$ 33.4                   | 178.3 $\pm$ 32.9 | 143.0 $\pm$ 20.5    | 134.4 $\pm$ 29.3                   | 19.9 $\pm$ 26.8                  | <b>246.3</b>   | 249.5     |
| Rev/EFC (£)                           | <b>1937 <math>\pm</math> 553</b>   | 1,960 $\pm$ 545  | 1806 $\pm$ 463      | 994 $\pm$ 737                      | 774 $\pm$ 385 <sup>†</sup>       | −411           | 3178      |
| <b>Block B: Operational Behaviour</b> |                                    |                  |                     |                                    |                                  |                |           |
| DA Rev (%)                            | 85.6 $\pm$ 33.8                    | —                | 14.0 $\pm$ 33.8     | <b>−1.7 <math>\pm</math> 81.7</b>  | 431.7 $\pm$ 3682.3               | 0.0            | —         |
| ID Rev (%)                            | 14.4 $\pm$ 33.8                    | —                | 86.0 $\pm$ 33.8     | <b>101.7 <math>\pm</math> 81.7</b> | −371.7 $\pm$ 3677.6              | 100.0          | —         |
| Deg Cost (%)                          | <b>51.6 <math>\pm</math> 7.6</b>   | —                | 53.1 $\pm$ 5.9      | 69.4 $\pm$ 13.5                    | 69.0 $\pm$ 27.3                  | <b>125.9</b>   | 39.7      |
| Idle Fraction (%)                     | 71.1 $\pm$ 5.8                     | 75.5 $\pm$ 5.3   | 82.5 $\pm$ 2.7      | 81.1 $\pm$ 4.3                     | <b>97.2 <math>\pm</math> 3.6</b> | 78.1           | —         |
| DA Plan Util (%)                      | <b>52.3 <math>\pm</math> 4.4</b>   | 57.5 $\pm$ 4.1   | 70.3 $\pm$ 2.6      | 71.3 $\pm$ 4.0                     | 93.2 $\pm$ 4.6                   | 78.1           | —         |
| Carbon displaced (tCO <sub>2</sub> )  | −4793.7 $\pm$ 745.3                | −4,740 $\pm$ 727 | −3761.3 $\pm$ 394.1 | −3618.3 $\pm$ 658.1                | −468.7 $\pm$ 575.1               | <b>−6436.8</b> | −6766.6   |

<sup>†</sup>Rev/EFC, DA Rev (%) and ID Rev (%) are unreliable for DQN given near-zero net profit (EFC  $\approx$  19.9). Do not interpret comparatively.

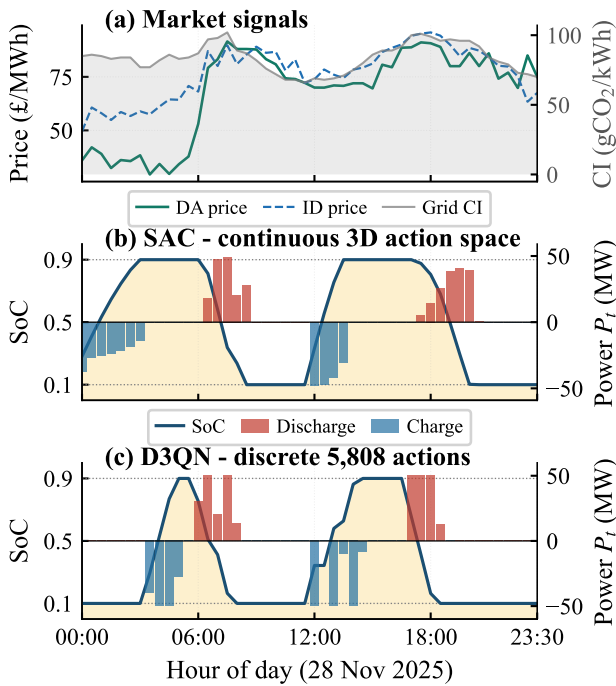


Figure 6: Single-day dispatch case study on 28 November 2025 at  $\lambda_{CI} = 0.1$ . (a) Market signals: DA and ID prices, with grid CI on the right axis. (b, c) SoC trajectory and applied power  $P_t$  for SAC and D3QN respectively.

### 5.3 Cycle Efficiency

The Rev/EFC hierarchy (Table 4) shows D3QN cycling only 6% more than DDQN yet generating 81% more revenue per cycle—a timing-quality rather than volume deficiency. DDQN’s elevated degradation share at fixed  $\kappa = \text{£}10/\text{MWh}$  indicates a learnable rather than structural failure: the same 5,808-action space supports D3QN’s superior timing once the overestimation bias is removed by the dueling factorisation.

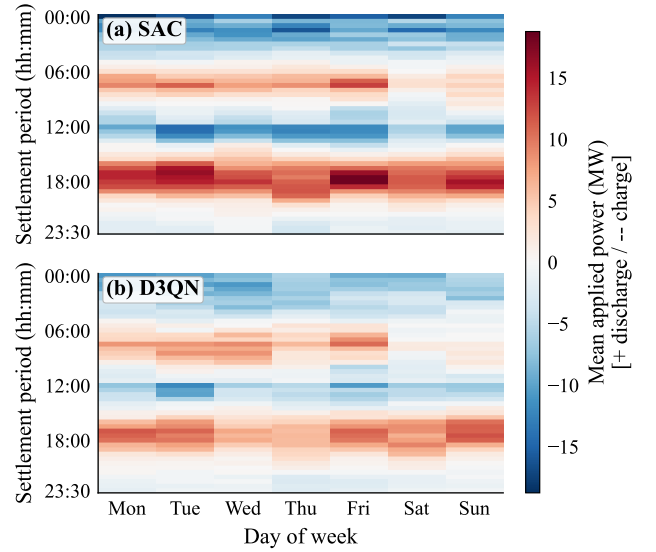


Figure 7: Mean dispatch power (MW) by settlement period (00:00–23:30, rows) and day of week (columns), five seeds pooled at  $\lambda_{CI} = 0.1$ . Red: mean discharge; blue: mean charge.

### 5.4 Ablations and Sensitivity Analysis

Figure 8 shows the profit–carbon trade-off for five  $\lambda_{CI}$  values (five seeds each). The Pareto sweep is non-monotonic:  $\lambda_{CI} = 0.1$  outperforms the carbon-agnostic baseline ( $\lambda_{CI} = 0.0$ ) by £16.9k. A plausible mechanism is the documented correlation in the GB grid between high carbon intensity and high demand [29], as gas-peaker dispatch raises both CI and DA prices.

Higher carbon penalties ( $\lambda_{CI} \geq 0.3$ ) destabilise SAC as profit variance rises sharply (Figure 8), consistent with conflicting DA-heavy and ID-positive modes across seeds.

As  $\lambda_{CI}$  increases, carbon displacement rises but cycling inflates (Figure 8); at  $\lambda_{CI} = 1.0$ , £476 k of phys-

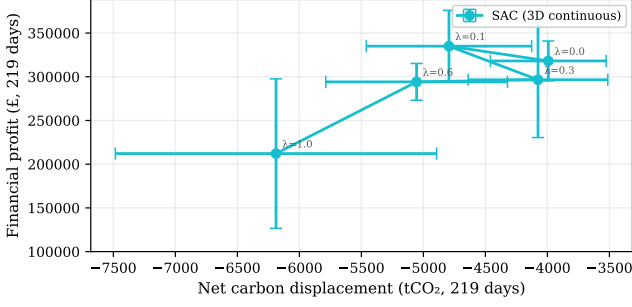


Figure 8: Profit-carbon Pareto frontier across carbon penalty weights  $\lambda_{CI} \in \{0.0, 0.1, 0.3, 0.6, 1.0\}$ . Error bars show  $\pm 1\sigma$  across five seeds.

ical wear yields only £212 k of profit. The LP oracle’s Pareto frontier confirms that beyond  $\lambda_{CI} \approx 0.1$ , carbon-for-cycle-life trading is strictly financially suboptimal.

## 5.5 Discussion and Limitations

The results demonstrate that structured action-space design and carbon-aware reward shaping are both effective and complementary levers for BESS dispatch, with the disc-11 ablation providing clean experimental evidence for the parameterisation’s structural contribution. Three interpretations follow, each qualified by a corresponding limitation.

*Action space versus algorithm architecture.* The successive profit uplifts within the discrete space (DQN  $\rightarrow$  DDQN  $\rightarrow$  D3QN, £8.6k  $\rightarrow$  £118k  $\rightarrow$  £251k) confirm that resolving overestimation bias and credit-assignment sparsity are both necessary for large-scale BESS dispatch [28, 27]; notably, the DDQN $\rightarrow$ D3QN step (£133k absolute) is the largest single algorithmic gain. The subsequent D3QN $\rightarrow$ SAC gap (£84k) is smaller in absolute terms but crosses a formulation boundary—from flat discrete to rank-based continuous parameterisation.

*Decomposing the SAC–D3QN gap.* The £84k performance gap (exceeding the combined seed-level standard deviations of both agents) was previously confounded across parameterisation, algorithm family, and power granularity. The SAC (disc-11) ablation resolves this: snapping SAC’s continuous outputs to 11 discrete levels per dimension at evaluation—matching D3QN’s action granularity—yields £335.5k  $\pm$  £41.8k, within noise of continuous SAC (£335.0k  $\pm$  £45.8k). The granularity contribution is  $-\text{£}0.5\text{k}$  ( $\approx 0\%$  of the gap); the entire £84k advantage is attributable to the rank-based parameterisation and the policy it enabled. This result establishes that SAC’s performance advantage is structural—it persists even when the continuous precision advan-

tage is removed—and is not an artefact of action-space resolution. At  $\lambda_{CI} \geq 0.3$ , seed variance rises sharply; carbon-profit conclusions beyond  $\lambda_{CI} = 0.1$  should be treated as indicative.

*Parameterisation and modelling constraints.* The disc-11 ablation establishes that granularity is negligible ( $\approx 0\%$  of the gap), leaving only the algorithm-family contribution (SAC (entropy-regularised actor-critic) vs the DQN family (value-based Q-learning)) as a residual confound. The sequential DQN $\rightarrow$ DDQN $\rightarrow$ D3QN ablation provides indirect evidence that algorithmic improvements within the DQN family yield diminishing returns once both overestimation and credit-assignment are resolved, suggesting the remaining gap is structural rather than algorithmic; a D3QN trained on the 3D parameterised space would confirm this definitively and is the natural next experiment. Beyond this, the rank-based expansion imposes symmetric activation and uniform slot power, structurally under-serving asymmetric price-spread days; a hierarchical parameterisation would address this.

## 6 Sustainability Analysis

*Embodied carbon payback.* The critical question is not whether BESS displaces carbon during operation, but whether that displacement is sufficient to offset the manufacturing burden. LFP cells carry an embodied footprint of 60–100 kgCO<sub>2</sub>/kWh [33]; for the 99.9 MWh system, cell-level embodied carbon is approximately 5,994–9,990 tCO<sub>2</sub>. Including Balance of System (BOS) components — inverters, cabling, HVAC, civil works — adds a further 30–50%, yielding a total embodied footprint of 7,792–14,985 tCO<sub>2</sub>. The operational displacement observed under the  $\lambda_{CI} = 0.1$  agent ( $-\text{4,793 tCO}_2$  over 219 days) annualises to  $\sim -7,988 \text{ tCO}_2/\text{year}$ , equivalent to removing  **$\sim 1,736$  gasoline passenger cars** from the road for one year [34]. This gives:

$$T_{\text{payback}} = \frac{7,792-14,985 \text{ tCO}_2}{7,988 \text{ tCO}_2 \text{ yr}^{-1}} = 0.98-1.87 \text{ years} \quad (20)$$

This sub-2-year payback against a 10–15 year asset life appears favourable, but three caveats temper the conclusion. First, using the average emission factor (AEF) underestimates marginal dispatch impact at scale [29, 35]; MEF would raise the payback. Second, the estimate assumes test-period performance extends year-round, whereas the window is a post-crisis regime with lower spreads. Third, even over 15 years the infrastructure

contributes 519–1,498 tCO<sub>2</sub>/year, which is not negligible relative to the  $\approx 7,988$  tCO<sub>2</sub>/year operational benefit.

*The carbon cost of profit-first dispatch.* At  $\lambda_{CI} = 0.0$  (pure profit), the agent displaces only  $-3,992$  tCO<sub>2</sub> over the test period, approximately 20% less than at  $\lambda_{CI} = 0.1$  ( $-4,793$  tCO<sub>2</sub>). Seed-level analysis shows that  $\lambda_{CI} = 0.0$  agents sometimes execute DA-heavy strategies that charge during periods of high price and high CI, approaching carbon-neutral or carbon-positive operation. The difference between  $\lambda_{CI} = 0.0$  and  $\lambda_{CI} = 0.1$  is  $-802$  tCO<sub>2</sub> over 219 days, yet the  $\lambda_{CI} = 0.1$  agent earns £16.9k more, so carbon-agnostic dispatch is both financially suboptimal and environmentally worse.

*Systemic scaling implications.* If the projected GB BESS deployment under the Clean Power 2030 pathway [1] were to adopt carbon-agnostic, profit-maximising algorithms universally, the aggregate carbon outcome would depend on how frequently price minima coincide with carbon-intensive periods. Where low-price charging windows are carbon-intensive — as occurs during gas-peaker dispatch in the UK — large-scale synchronised charging shifts demand towards dirtier marginal generation. This is a systemic risk that does not appear at the individual asset level. Carbon-aware regularisation ( $\lambda_{CI} > 0$ ) is therefore not merely an operator preference; at fleet scale, it may be a necessary safeguard against BESS collectively worsening the emissions profile they are deployed to improve. We accordingly recommend that fleet operators above a 1 GW aggregate-capacity threshold adopt  $\lambda_{CI} \geq 0.1$  as a default operating point, and that system operators consider this threshold when designing future market-coupling or carbon-attribution mechanisms.

## 7 Conclusions

This work shows that *how* a joint day-ahead and intraday scheduling problem is formulated matters more than *which* algorithm solves it. By encoding market-structural knowledge directly into a three-dimensional action parameterisation, the agent is freed from discovering the ranked structure of wholesale prices through interaction alone—a task that proves intractable at the 49-dimensional slot-wise scale. An evaluation-time action-discretisation ablation confirms that the resulting performance advantage is entirely structural: the policy retains its superiority even when continuous precision is removed, ruling out action granularity as a con-

found. The co-optimisation of carbon alongside profit further reveals that sustainability and commercial objectives are not in tension in the GB wholesale market; modest carbon-aware regularisation is simultaneously the financially and environmentally superior operating strategy. Together, these findings establish a reusable design principle for RL in markets with contract structure: embedding domain priors into the action space, rather than relying on the algorithm to discover them, is the higher-leverage engineering decision.

The principal limitation is that the algorithm-family contribution—the difference between SAC’s off-policy entropy-regularised learning and a DQN-family agent trained on the same parameterised space—remains unquantified. A D3QN trained on the 3D formulation would resolve this and is the natural next experiment. More broadly, the symmetric rank-based parameterisation structurally under-serves asymmetric or spike-dominated price profiles; a hierarchical extension separating commitment shape from aggressiveness would improve expressivity. At fleet scale, the carbon accounting relies on average rather than marginal emission factors, understating displacement impact; transitioning to marginal emission models and incorporating a rolling-horizon benchmark with realistic forecasts remain the two open extensions most relevant to commercial deployment.

## 8 Declarations

### 8.1 Acknowledgements

I, Giacomo Fiorani, confirm that the work presented in this report is my own. Where information has been derived from other sources, I confirm that this has been indicated in the report. I thank Dr Peng Wu for supervision and guidance throughout this project.

### 8.2 Use of AI

Claude Sonnet 4.6 (Anthropic) was used in an assistive role to proofread author-written text for grammar, spelling, and clarity.

### 8.3 Code Availability

The project code is available on GitHub on request from the author.

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